# SRPCNet: Self-Reinforcing Perception Coordination Network for Seamless Steel Pipes Internal Surface Defect Detection

Hongshu Chen, Kechen Song, *Member, IEEE*, Wenqi Cui, Tianle Zhang, Yunhui Yan, and Jun Li

***Abstract*—Seamless steel pipes (SSPs) are vital material for industries. However, internal surface defects (ISDs) in SSPs are challenging to detect, and will significantly affect SSPs performance and lifespan. Existing detection methods are labor-intensive and have low visualization of detection results. Therefore, this article present a novel detection system comprising the Pipeline All-aspect internal Surface defect Spiral detecting robot and an interactive visualization software. After testing in the SSPs factory, the system achieves comprehensive, wireless and efficient detection and visualization for ISDs. In addition, we construct a dataset for ISDs in SSPs, named as SSP2000. The dataset contains 2000 images across nine defect categories, with many challenges in it. Furthermore, to accurately detect defects, we design the SRPCNet which can effectively address the challenges. Specifically, we first use the synergize perception augmentation module to enrich the feature space and to enhance the perception. Then, the hierarchical attention integrate module merges deep and shallow features using adaptive attention weights. Finally, the bilateral self-fusion module fully exploits intralayer features and produce prediction results. The proposed SRPCNet outperforms existing methods on eight evaluation metrics.**

***Index Terms*—Detection system, pipe internal surface dataset, salient object detection (SOD), seamless steel pipes (SSPs), surface defect detection.**

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The source code and the dataset are publicly available at https://github.com/VDT-2048/SRPCNet.

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## I. Introduction

SEAMLESS steel pipes (SSPs) are important industrial raw materials with significant applications in petrochemical industry, aerospace, and high-pressure equipment manufacturing [1]. Fig. 1 shows the manufacturing process of SSPs in a real industrial scenario. Different from welded steel pipes, which are formed by spiral or straight welding of strip steel, SSPs are primarily formed from pipe billets through processes, such as pickling, cold drawing (or hot rolling), and grinding. Compared with welded steel pipes, which are prone to internal defects in the weld seam, SSPs are more likely to develop surface defects such as cracks, scratches, pits, and scale due to the influence of raw materials and production processes. These defects can seriously affect the performance and lifespan of the SSPs [2]. For weld seam defects in welded steel pipes, there are many detection methods based on X-ray images [3], [4], [5], [6]. For external surface defects of SSPs, which are easier to be captured, there are also some detection methods [7], [8] have been developed. However, internal surface defects (ISDs) are harder to be observed or detected due to their enclosed environment. Therefore, detecting ISDs in SSPs is critical and challenging.

Existing methods for detecting ISDs can be classified into nondestructive testing (NDT) methods and visual detection methods. NDT methods utilize specialized equipment to convert defects into electrical or waveform signals, including eddy current testing [9], ultrasonic testing [10], and magnetic flux leakage testing [11], etc. Specifically, eddy current testing is an inspection method that is easily automated but is only suitable for detecting surface and near-surface defects in conductive materials. Ultrasonic testing can accurately detect internal defects, such as pores and cracks, but it is susceptible to noise interference. Magnetic flux leakage testing is effective in accurately detecting small surface defects, but it is limited to inspecting ferromagnetic materials [12]. Visual inspection methods are intuitive and easy to operate, mainly including manual inspection using industrial endoscopes and inspection by pipeline robots based on machine vision. As shown in Fig. 1, although manual inspection is simple to operate, it is labor-intensive, subjective and has the risk of missing defects. In addition, during the manual grinding phase, workers use grinding tools to repair ISDs in semifinished SSPs, which means the results from the detection stage, such as the location and size of ISDs, are crucial for guiding the grinding process. However, after observing in the factory, it was found that there is a lack of effective communication between the two stages, causing grinding workers need to estimate the approximate location of defects again through visual observation,

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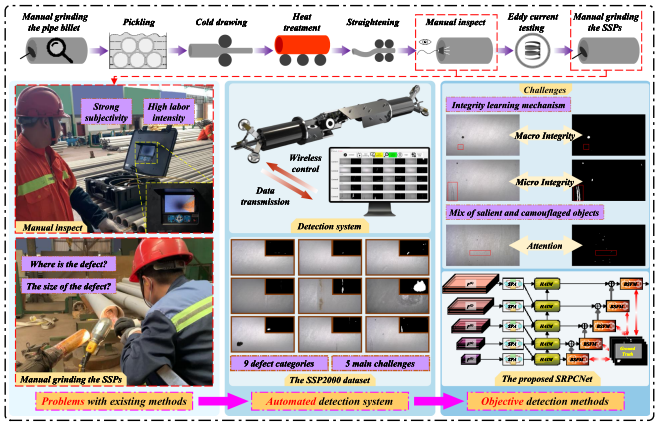



Fig. 1. Problems with existing methods for ISDs detection in the manufacturing process of SSPs, and proposed detection system and method.

thereby reducing production efficiency. Pipeline robots based on machine vision can meet high precision requirements [13], [14], but most existing pipeline robots are wired, which means the robot needs to enter and exit from the same end of the pipeline, which undoubtedly reduces the overall detection efficiency.

To address the challenges of detecting ISDs in SSPs and achieve accurate, efficient, and visualized nondestructive detecting, we propose a comprehensive detection system. The system includes a Pipeline All-aspect internal Surface defect Spiral detecting robot (PASS) and an interactive visualization software (IVS) deployed on the operating terminal. Specifically, the PASS moves through the SSPs, wirelessly controlled by the IVS to ensure comprehensive image collection of the internal surface. The captured images are then wirelessly transmitted to the operation terminal to be analyzed. The detection results are visually displayed within the IVS. The solution not only enables continuous inspection of the piled-up SSPs, but also provides visualized results that can aid in subsequent manual grinding processes. Practical testing in an SSPs factory demonstrates that the detection system can effectively complete the task of detecting ISDs in SSPs.

An ISDs dataset for SSPs is necessary to validate the detection capability of the proposed detection system. However, to the best of our knowledge, no such dataset currently exists for further research. To verify the capability of the system, we use the PASS to collect internal surface images from a large number of cold-drawn SSPs in the factory. We manually remove all normal images from collected images under the guidance of professional quality control workers and obtain 2000 images, constructing the SSP2000 dataset, which covers nine defect categories, with quantities reflecting actual distribution. Moreover, to ensure the applicability of the detection system in real-world production, we do not perform any cropping or preprocessing on the images in the SSP2000 dataset. We also identify and analyze five categories of challenges and nine subcategories in the SSP2000 dataset.

The salient object detection (SOD) methods have achieved notable success in the field of defect detection [15], [16], [17]. However, due to the challenges in the uncropped internal surface images of SSPs, such as cross-scale multiple objects (CMOs) and the mix of camouflaged and noncamouflaged objects (MCNOs), these methods fail to detect adequately. To better detect defects in the internal surface images of SSPs, we think that the following two problems must be fully considered.

1) How to satisfy the integrity learning mechanism [18]? This includes the macroscopic integrity in multiobject detection in an image and the microscopic integrity in the detection of discrete or large-scale defects.

2) How to pay enough attention to camouflaged targets when both salient and camouflaged targets coexist in an image?

Based on the above-mentioned considerations, we propose the self-reinforcing perception coordination network (SRPCNet). To achieve robust perception capability and enhance feature representation, we first implement synergize perception augmentation (SPA). Next, we employ the hierarchical attention integrate module (HAIM) to fully interact the detailed and semantic information across layers, thereby enhancing the perception of long-range feature information and improving the detection integrity. Finally, to thoroughly extract information from the fused features, we utilize the bilateral self-fusion module (BSFM) to self-reinforce single-level features, followed by supervised upsampling and cascade operations for each feature level.

In summary, the main contributions of this article are summarized as follows.

1) We propose a novel detection system for ISDs in SSPs, consisting of the PASS robot and the IVS. The system is capable of accurate and efficient detection for ISDs in SSPs with internal diameters ranging from 100 to 150 mm, with the detection efficiency of 6 m/min.
2) To the best of our knowledge, we propose the first dataset of ISDs in the SSPs (SSP2000). The SSP2000 dataset contains nine categories of common ISDs in SSPs, as well as five categories of challenges and ten subcategories.
3) An SRPCNet is proposed, which can detect surface defect efficiently. We compare SRPCNet with 15 state-of-the-art methods and SRPCNet achieves the best detection results on eight evaluation metrics used.

## II. Related Works

### A. SOD Methods for Steel Surface Defects

Steel is a crucial material for industrial development, and detecting defects on steel surface is essential. In recent years, many researchers regarded steel surface defects as salient objects for detection. In the field of SOD methods for strip steel defects, Song et al. [15] proposed EDRNet, which achieved excellent detection results by using a fully convolutional neural network and attention mechanisms during the encoding stage. Ding et al. [16] focused on the edge information of strip steel defects and proposed CSEPNet, which effectively calibrated adjacent features of different scales by using a cross-scale calibration module and a cross-scale feature interweaving module. For the SOD methods of steel rail defects, Niu et al. [19] constructed the RSDDS-113 steel rail dataset and proposed an unsupervised stereo saliency detection method based on a binocular line scan system. Huang et al. [20] were the first to use skeletons to assist in defect localization for defect detection. They designed SA2F, which is compatible with both RGB and RGBD image detection, by using a hierarchical visual perception module and a stereoscopic attention mechanism. Camouflaged object detection (COD) also has made lots of research progress [21], [22], [23]. Some steel surface defects share similar characteristics with COs, such as

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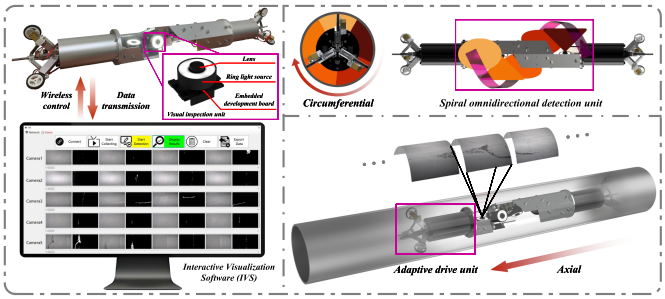



Fig. 2. Proposed detection system for ISDs in SSPs and the detection principle of the PASS.

low contrast and blurred boundaries. Luo et al. [24] proposed a CDDNet for camouflaged defects on steel surfaces. The designed IDTEM enhances uncertain boundaries and deceptive textures in images, and the effectiveness of their method was tested and validated on multiple datasets. However, since most existing steel surface defect datasets are made of cropped images, it is rare for a single image to contain both camouflaged and non-COs simultaneously, which could affect the attention of the network. Therefore, the detection of mixed camouflaged and non-COs remains further research.

### B. Detection of Surface Defects for SSPs

Cold drawing and hot rolling are primary methods for manufacturing SSPs. SSPs often develop surface defects due to the influence of raw materials and production processes. The detection of surface defects in SSPs is mainly categorized into external surface detection and internal surface detection. Commonly used detection methods include visual inspection, eddy current testing, and ultrasonic testing. Although these methods provide relatively effective results, they still face challenges, such as high labor intensity, susceptibility to interference, and a low level of visualization [2]. With the advancement of deep learning, methods utilizing machine vision for surface quality inspection continue emerging [2], [7], [8]. However, these methods mainly focused on external surface defects where is relatively easy to set up a vision inspection system. Suffering from the spatial constraints inside the SSPs, ISDs detection faces significant challenges. What is more, within the scope of our investigation, there is currently no ISDs dataset for SSPs, which greatly hinders the research on visual inspection methods for ISDs. Based on the above-mentioned discussion, this article aims to address these problems by constructing a comprehensive visual inspection system for detecting ISDs in SSPs. In addition, we construct the SSP2000 dataset of ISDs in SSPs by using real internal pipe surface images, and designed an SRPCNet for detection.

## III. Proposed System and Dataset

### A. Defect Detection System

To address the problems of high labor intensity, subjectivity, and low visualization levels in existing methods for detecting ISDs in SSPs, we propose a novel vision-based detection system for ISDs in SSPs. As shown in Fig. 2, the proposed detection system comprises a PASS and an IVS.

The PASS consists of a spiral omnidirectional detection unit and two adaptive drive units. The spiral omnidirectional detection unit is composed of five visual detection units; each includes an M12 industrial lens, a ring light source, and an embedded development board for image capturing and transmission. The adaptive drive unit is designed to adapt the PASS to different pipe diameters, supply electricity, and drive the PASS forward. Fig. 2 also illustrates the detection principle of the PASS, each of the five visual inspection units faces a different direction of the inner wall, enabling exhaustive detection in the circumferential direction. Meanwhile, by coordinating the speed of the PASS and the image capturing speed, each image captured by a single unit overlaps slightly with the previous one along the length of pipes, ensuring exhaustive detection in the axial direction.

The IVS has function of wirelessly control the visual detection unit (involving network connection, starting or stopping image capturing, and high-speed data transmission to the server) and visual display of detection results. When the PASS enters the pipe, the operator controls visual detection units by the IVS to initiate image capturing. After the PASS passes through the pipe, the image capturing stops, and the operator then controls visual detection units via the IVS to send the captured images to the server for detection. The visualized results of the detection will be displayed within the IVS. The detection system achieves objective, intuitive, and efficient detection of the challenging ISDs in SSPs. Furthermore, the detection results can be used to guide the grinding stage after detection in the production of the cold-drawing SSPs.

### B. Dataset Construction Process

The images of the SSP2000 are collected by the PASS in a cold-drawing SSPs factory. Due to the low defect rate in the production process, the PASS has to collect images from a large number of SSPs, then samples with defects are manually selected. Considering the characteristics of ISDs, such as blurry edges and varying shapes, we invite professional SSPs quality inspectors to guide and supervise the selection process. Ultimately, 2000 defect images are selected for the dataset construction. Notably, aside from discarding some low-quality defect images, we use nearly all the collected defect images, ensuring the data distribution of each defect category reflects real conditions. To meet the detection principle of the PASS, we preserve the original image size ($640 \times 360$) and use photoshop for pixel-level annotation for each defect image.

We divide the 2000 defect images into nine categories, as shown in Fig. 3, which are cross-striation, peeling, pit, pockmark, rust, sandstreak, scale, scratch, and multiclass, under the guidance of quality inspection workers. To ensure the distribution of defect categories in the training and testing sets aligns with real conditions, we randomly select images from each category in a 7:3 ratio to construct the training and testing sets, respectively, resulting in 1400 images for the training set and 600 images for the testing set.

### C. Challenges in the SSP2000

Steel surface defects are characterized by irregular shapes and poor edge textures, along with the specific characteristics of ISDs in SSPs, we identify five main categories of challenges in the SSP2000 dataset, including ten subcategories. As shown in Fig. 4, the five main challenges include the object challenge,

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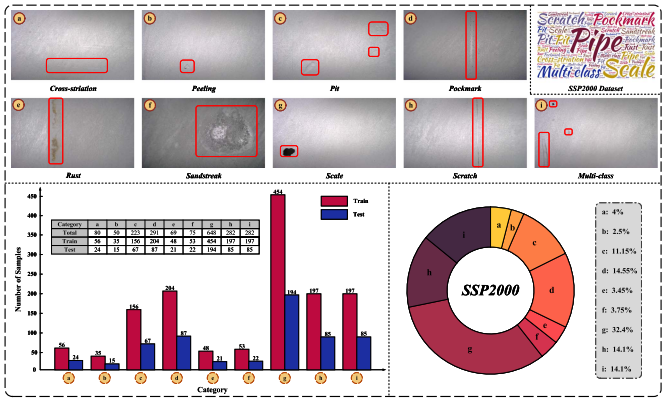



Fig. 3. Sample gallery of the SSP2000 dataset and the distribution of each defect type in the training and testing sets. Please zoom in for the best view.

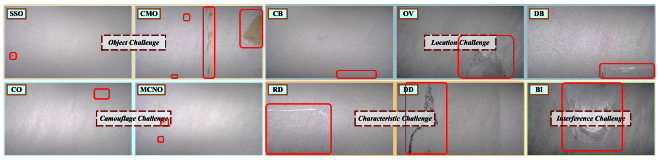



Fig. 4. Challenges in the SSP2000. Please zoom in for the best view.

TABLE I
CHALLENGE DISTRIBUTION IN THE SSP2000 DATASET

| Challenges | SSO | CMO | CO | MCNO | CB | OV | DB | RD | DD | BI |
|---|---|---|---|---|---|---|---|---|---|---|
| SSO | 1389 | 266 | 317 | 135 | 480 | 251 | 188 | 184 | 800 | 72 |
| CMO | 266 | 289 | 36 | 31 | 121 | 151 | 67 | 57 | 204 | 7 |
| CO | 317 | 36 | 495 | 162 | 134 | 62 | 48 | 38 | 200 | 23 |
| MCNO | 135 | 31 | 162 | 162 | 49 | 34 | 30 | 34 | 99 | 9 |
| CB | 480 | 121 | 134 | 49 | 663 | 169 | 190 | 83 | 243 | 44 |
| OV | 251 | 151 | 62 | 34 | 169 | 471 | 139 | 171 | 271 | 16 |
| DB | 188 | 67 | 48 | 30 | 190 | 139 | 295 | 112 | 147 | 13 |
| RD | 184 | 57 | 38 | 34 | 83 | 171 | 112 | 353 | 195 | 17 |
| DD | 800 | 204 | 200 | 99 | 243 | 271 | 147 | 195 | 922 | 25 |
| BI | 72 | 7 | 23 | 9 | 44 | 16 | 13 | 17 | 25 | 114 |

the camouflage challenge, the location challenge, the characteristic challenge, and the interference challenge. The object challenge includes small salient objects (SSOs) and CMOs. The camouflage challenge includesCOs and the MCNO. The location challenge includes center bias (CB), out of view (OV), and defocused blurring (DB). The characteristic challenge includes reflective defects (RDs) and discrete defects (DDs). The interference challenge includes background interference (BI). Table I shows the challenge distribution and Fig. 5 shows the challenge distribution in the training and testing sets.

Influenced by the production process of SSPs and various causes of defects, the size of defects varies significantly, increasing the detection difficulty. In addition, to align with the detection principles of the PASS, we do not crop the images, which results in each image potentially containing multiple defect objects and multiple challenges, further complicating the detection. The relationship between the various challenges is more intuitively shown in Fig. 6(a) and (b).

### D. Advantage Analysis

The SSP2000 dataset has significant differences and advantages compared with existing pipeline datasets.

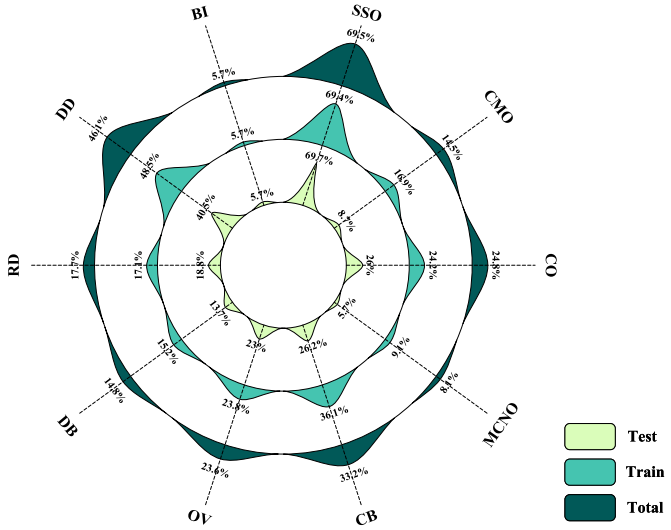



Fig. 5. Challenge distribution in the training and testing sets.

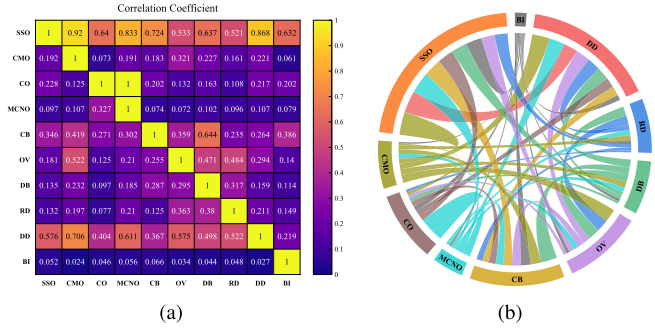



Fig. 6. (a) Heat map and (b) chord diagram of challenges distribution.

1) Unlike existing axially captured pipeline datasets, the SSP2000 dataset is the first dataset for ISDs in SSPs, which has the radial view of the internal surface. The SSP2000 dataset fills the gap of ISDs images in SSPs, which are difficult to collect.
2) The images in the SSP2000 dataset are real images collected by the PASS in an SSPs factory. The images are not cropped, denoised, or undergone other data enhancement operations. Each image is annotated at the pixel level under the guidance of quality inspection workers, and the data distribution of each defect category aligns with actual conditions. Therefore, the SSP2000 dataset accurately reflects the real conditions of ISDs in SSPs.
3) The SSP2000 dataset includes a large number of defect images and encompasses various challenges, making it both challenging and difficult for detection.

## IV. PROPOSED METHOD

### A. Overview

Accurately detecting defects on the internal surface of SSPs is a challenging task. Therefore, we propose an SRPCNet to detect, as shown in Fig. 7. We use VGG16 as the backbone to

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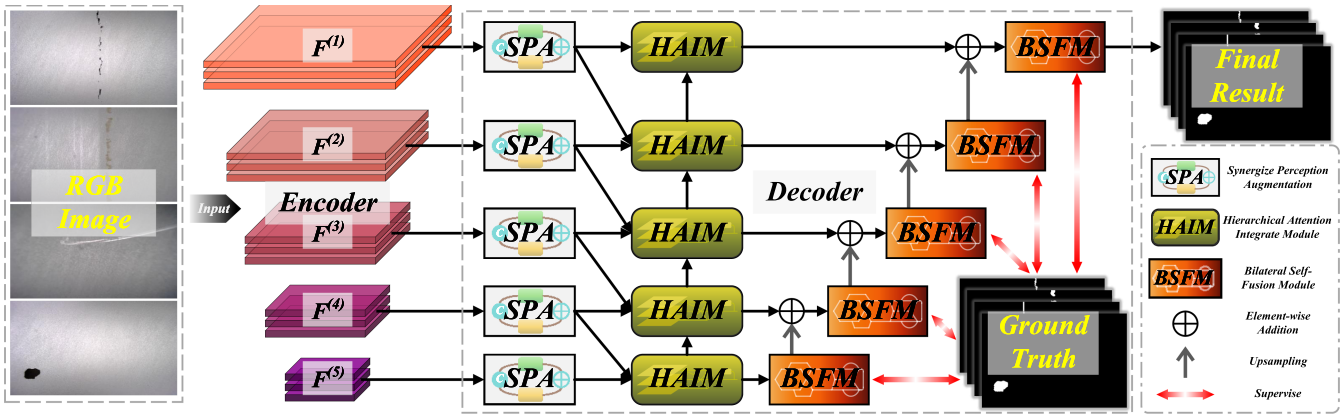



Fig. 7. Framework of the proposed SRPCNet.

extract multilevel features. To obtain sufficient information, we retain all feature layers obtained by the backbone, denoted as $F^{(i)}(i = 1, 2, 3, 4, 5)$. After extracting the features, the SPA is first utilized to enrich the feature space and enhance the perception capabilities of the method. Then, the HAIM integrates deep and shallow features through different attention weights, merging detailed and semantic information. Finally, the BSFM is used to fully exploit the useful information within the fused intralayer features. By supervising each layer of features, a saliency prediction map is finally generated. In addition, it is worth mentioning that there is no parameter sharing between identical modules.

## B. Synergize Perception Augmentation

In SSP2000 dataset, there are challenges, such as CMO and MCNO, in many images, which require the detection method to have sufficient perceptive and expressive capabilities. According to previous researches [25], [26], hybrid dilated convolution can effectively expand the receptive field while maintaining the information continuity, and asymmetric convolution kernels can enhance feature representation. Therefore, to achieve both a large receptive field and the enhanced feature expression, we propose the SPA, as shown in Fig. 8. Given the input features $F^{(i)}$, we first use the $1 \times 1$ convolution operation to obtain features $F_l^{(i)}$ with uniform channels, then we use the asymmetric depthwise separable conv block (ADSCB) and the hybrid dilated conv block to enhance features and expand the receptive field as follows:

$$F_{\text{spa}}^{(i)} = F_l^{(i)} + F_a^{(i)} + F_h^{(i)} \tag{1}$$

where $F_{\text{spa}}^{(i)}$ is features produced by the SPA. $F_a^{(i)}$ and $F_h^{(i)}$ are features produced by the ADSCB and HACB, respectively.

*1) Asymmetric Depthwise Separable Conv Block:* We integrate the asymmetric convolution operation and depthwise separable convolution operation, along with a squeeze-and-excitation block [27], which is used to obtain channel attention weights, enhancing the feature expression. Given the input feature $F_l^{(i)}$, the process can be formulated as follows:

$$F_a^{(i)} = \Sigma_{\Delta=3\times3,1\times3,3\times1} \hat{f}_{1\times1}[S(f_\Delta(F_l^{(i)}))] \tag{2}$$

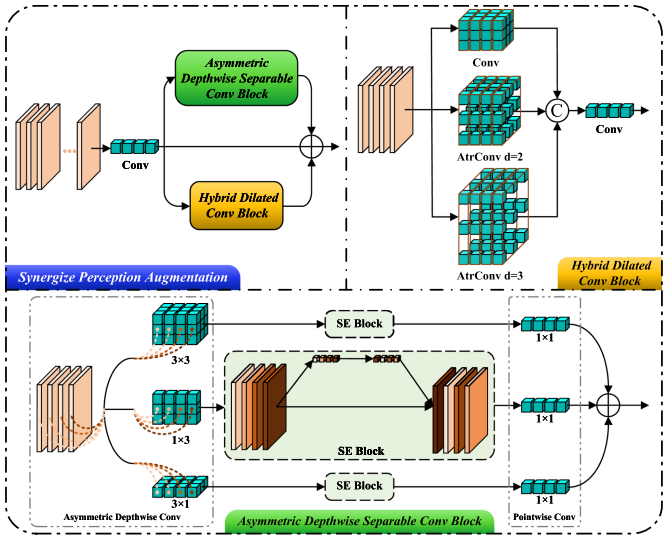



Fig. 8. Structure of the SPA.

where $f_\Delta(*)$ is the depthwise convolution operation, and $\Delta = 3 \times 3, 1 \times 3, 3 \times 1$ means the kernel size of asymmetric convolution operation. $\hat{f}_{1\times1}[*]$ is the pointwise convolution operation. $S(*)$ represents the SEBlock, which can be formulated as follows:

$$S(F) = F \odot (\text{sigmoid}(\text{Conv}(R(\text{Conv}(\text{Avg}(F)))))) \tag{3}$$

where $\text{Avg}(*)$ is the average pooling operation, $R(*)$ is the relu operation, and $\odot$ is the elementwise multiplication. Specifically, the SEBlock generates channel attention weights through a structure composed of a global average pooling layer, two fully connected layers, and two activation functions. By multiplying the obtained weights with the input feature, the block selectively emphasizes certain channel features, thereby enhancing the feature representation.

*2) Hybrid Dilated Conv Block:* By reasonably using a combination of dilated convolution operations with different dilation rates, we enhance the receptive field of the model while ensuring the continuity of feature information, the process can be formulated as follows:

$$F_h^{(i)} = \text{Conv}_{1\times1}[\text{Concat}(\text{DConv}_d(F_l^{(i)}))], d = 1, 2, 3 \tag{4}$$

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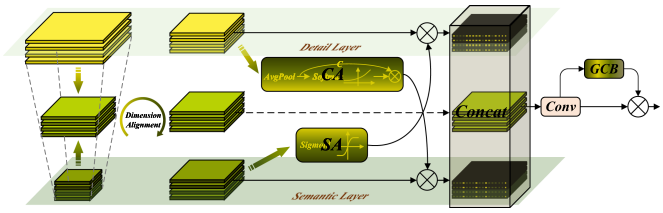



Fig. 9. Structure of the HAIM.

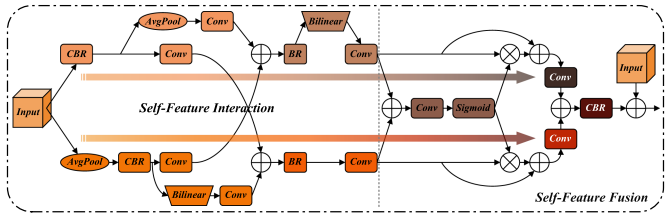



Fig. 10. Structure of the BSFM.

where $\mathrm{DConv}_d(*)$ is the dilated convolution operation and $d$ is the dilated rate. $\mathrm{Concat}(*)$ refers to the concat function, which is used to concatenate two or more features along a specified dimension.

### C. Hierarchical Attention Integrate Module

Detecting large-scale defects and DDs within a single image is another challenge faced by the method. Therefore, after obtaining the enhanced multilayer features through SPA, we design the HAIM, as shown in Fig. 9, to ensure comprehensive detection of as many defects as possible within the entire image while completely detecting large-scale defects. Specifically, we first align the dimensions of the enhanced multilayer features. Then, we use different attention weights to perform semantic and detailed information interaction on cross-layer features. When there are not enough feature layers, such as the shallowest and deepest layer, we use adjacent features for interaction. To achieve comprehensive detection of DDs and large-scale defects, we use the global context block [28] to establish global contextual information and enhance the perception of long-distance features. It is important to note that while using HAIM, we fuse features from deep to shallow layers and use the fused features from the deeper layers as input for the shallow layers' fusion, thereby achieving comprehensive multilevel feature fusion. The specific process is as follows:

$$F_{\mathrm{haim}}^{(i)} = \begin{cases} \mathbb{M}_i[F_{\mathrm{spa}}^{(i)'}, F_{\mathrm{haim}}^{(i+1)}], & i = 1 \\ \mathbb{M}_i[F_{\mathrm{spa}}^{(i)'}, F_{\mathrm{haim}}^{(i+1)}, F_{\mathrm{spa}}^{(i-1)'}], & i = 2,3,4 \\ \mathbb{M}_i[F_{\mathrm{spa}}^{(i)'}, F_{\mathrm{spa}}^{(i-1)'}], & i = 5 \end{cases} \tag{5}$$

where $F_{\mathrm{haim}}^{(i)}$ is the feature produced by the HAIM, and $F_{\mathrm{spa}}^{(i)'}$ represents the multilayer features after the dimension alignment. $\mathbb{M}_i[*]$ means the specific calculate processes of the HAIM, which are formulated as follows:

$$\mathbb{M}_i[F, F_{dl}, F_{sl}] = \mathrm{Conv}_{1\times1}(\mathrm{Concat}(F_{dl} \odot \mathrm{SA}(F_{sl}), F, F_{sl} \odot \mathrm{CA}(F_{dl}))), i = 2,3,4 \tag{6}$$

$$\mathbb{M}_i[F_{dl}, F_{sl}] = \mathrm{Conv}_{1\times1}(\mathrm{Concat}(F_{dl} \odot \mathrm{SA}(F_{sl}), F_{sl} \odot \mathrm{CA}(F_{dl}))), i = 1,5 \tag{7}$$

where $F$, $F_{dl}$, and $F_{sl}$ represent the intermediate layer feature, the shallow detailed feature and the deep semantic feature, respectively. $\mathrm{CA}(*)$ and $\mathrm{SA}(*)$ are channel attention and spatial attention as follows:

$$\mathrm{CA}(F_{dl}) = \mathrm{softmax}(\mathrm{Avg}(F_{dl})) \odot \ell(\mathrm{Avg}(F_{dl})) \tag{8}$$

$$\mathrm{SA}(F_{sl}) = \mathrm{sigmoid}(F_{sl}) \tag{9}$$

where $\ell(*)$ is to obtain the channel number of the feature.

### D. Bilateral Self-Fusion Module

The local defocus blurring of ISDs in SSPs can affect detection accuracy. When defects are located in the defocused area, their edge resolution is extremely low, and both the background and defect textures attenuate, affecting the saliency of defects. Therefore, we design the BSFM to fully extract the information in features, as shown in Fig. 10. Given the features $F_{\mathrm{haim}}^{(i)}$ obtained from the HAIM, we perform downsampling and convolution operations to obtain features with changed dimensions, forming a bilateral structure. One side consists of original dimension features, and the other side consists of dimension-changed features. The two sides undergo self-feature interaction and self-feature fusion in sequence, enhancing the bilateral feature correlation while retaining the original information. Subsequently, we upsample the self-fused features and perform a deep-to-shallow cascade operation, as shown in the following process:

$$F_{\mathrm{sfi}}^{(i)}, \hat{F}_{\mathrm{sfi}}^{(i)} = \begin{cases} \mathbb{L}_{\mathrm{sfi}}(\mathrm{Up}(\mathrm{CBR}(F_{\mathrm{bsfm}}^{(i+1)})) + F_{\mathrm{haim}}^{(i)}) & \\ \qquad\qquad i = 1,2,3,4 & \\ \mathbb{L}_{\mathrm{sfi}}(F_{\mathrm{haim}}^{(i)}), \qquad i = 5 & \end{cases} \tag{10}$$

where $F_{\mathrm{sfi}}^{(i)}$ and $\hat{F}_{\mathrm{sfi}}^{(i)}$) are original dimension features and dimension-changed features produced by the self-feature interaction. $\mathbb{L}_{\mathrm{sfi}}(*)$ is the self-feature interaction operation, and $\mathrm{Up}(*)$ is the upsample operation. $\mathrm{CBR}(*)$ represents performing convolution, batch normalization, and relu operations in sequence

$$F_{\mathrm{bsfm}}^{(i)} = \mathbb{L}_{\mathrm{sff}}(F_{\mathrm{sfi}}^{(i)}, \hat{F}_{\mathrm{sfi}}^{(i)}) + F_{\mathrm{haim}}^{(i)} \tag{11}$$

where $F_{\mathrm{bsfm}}^{(i)}$ is the feature produced by the BSFM. $\mathbb{L}_{\mathrm{sff}}(*)$ represents the operation of self-feature fusion.

*1) Self-Feature Interaction:* By performing downsampling and convolution operations on the input features, dimension-changed features are obtained. Both the original dimension features and the dimension-changed features undergo convolution and attention operations sequentially to achieve feature interaction, thereby fully extracting the information from each layer of features, the process can be formulated as follows:

$$F_{\mathrm{sfi}}^{(i)'} = \mathrm{Conv}(\mathrm{CBR}_{hh}(F_{\mathrm{in}}^{(i)})) \tag{12}$$

$$\hat{F}_{\mathrm{sfi}}^{(i)'} = \mathrm{Conv}(\mathrm{CBR}_{hl}(\mathrm{Avg}(F_{\mathrm{in}}^{(i)}))) \tag{13}$$

where $F_{\mathrm{sfi}}^{(i)'}$ and $\hat{F}_{\mathrm{sfi}}^{(i)'}$ are the original dimension features and the dimension-changed features, respectively, and $F_{\mathrm{in}}^{(i)}$ is the input feature. The $hh$ and $hl$ represent convolution operations from high dimensions to high dimensions and from high dimensions

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to low dimensions, respectively. Then

$$F_{\text{sfi}}^{(i)} = \text{Conv}(\text{Up}(\text{BR}[\text{Conv}(\text{Avg}(\text{CBR}_{hh}(F_{\text{in}}^{(i)}))) + \hat{F}_{\text{sfi}}^{(i)'}])) \tag{14}$$

$$\hat{F}_{\text{sfi}}^{(i)} = \text{Conv}(\text{BR}[\text{Conv}(\text{Up}(\text{CBR}_{hl}(\text{Avg}(F_{\text{in}}^{(i)})))) + F_{\text{sfi}}^{(i)'}]) \tag{15}$$

where $\text{BR}(*)$ represents performing batch normalization and relu operations in sequence.

*2) Self-Feature Fusion:* After obtaining the bilaterally interacted features $F_{\text{sfi}}^{(i)}$ and $\hat{F}_{\text{sfi}}^{(i)}$, we fuse the bilateral features. First, calculate the attention weight as follows:

$$\alpha = \text{sigmoid}(\text{Conv}(F_{\text{sfi}}^{(i)} + \hat{F}_{\text{sfi}}^{(i)})) \tag{16}$$

where $\alpha$ is the attention weight of the bilateral features. Then, we utilize the attention weight to adaptively enhance the correlation of bilateral features as follows:

$$\begin{aligned} F_{\text{sff}}^{(i)} = \text{CBR}(&\text{Conv}(\alpha \odot F_{\text{sfi}}^{(i)} + F_{\text{sfi}}^{(i)}) \\ &+ \text{Conv}(\alpha \odot \hat{F}_{\text{sfi}}^{(i)} + \hat{F}_{\text{sfi}}^{(i)})) \end{aligned} \tag{17}$$

where $F_{\text{sff}}^{(i)}$ is the feature produced by the self-feature fusion.

### E. Supervision Strategy

To ensure compatibility with ground truth maps during both training and inference process, we adjust channels and dimensions of features to match ground truth maps. We use BCE loss and IoU loss for supervision, which is calculated as

$$L = L_{\text{BCE}} + L_{\text{IOU}} \tag{18}$$

where $L$ is the total loss. $L_{\text{BCE}}$ and $L_{\text{IOU}}$ are BCE loss and IOU loss. Specifically, the BCE loss is formulated as follows:

$$\begin{aligned} L_{\text{BCE}} = -\sum_{x=1}^{H}\sum_{y=1}^{W}&[G(x,y)\log(P(x,y)) \\ &+ (1-G(x,y))\log(1-P(x,y))] \end{aligned} \tag{19}$$

where $W$ and $H$ are the width and height of the images, respectively. $G(x,y)$ and $P(x,y)$ are the ground truth label and the predicted saliency label at the location $(x,y)$, respectively. The IOU loss is formulated as follows:

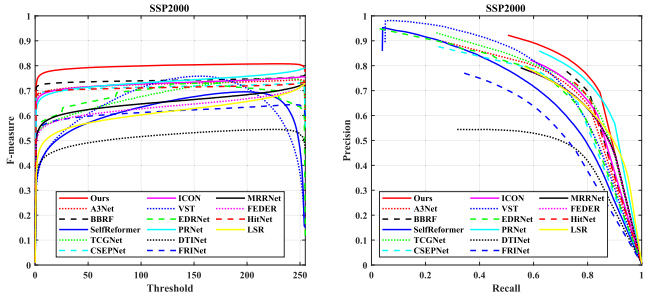
$$L_{\text{IOU}} = 1 - \frac{\sum_{x=1}^{H}\sum_{y=1}^{W} P(x,y)G(x,y)}{\sum_{x=1}^{H}\sum_{y=1}^{W}[P(x,y)+G(x,y)-P(x,y)G(x,y)]}. \tag{20}$$

Finally, we upsample the supervised features and perform a deep-to-shallow cascade operation to get the predicted map.

## V. Experiments

### A. Implementation Details

Our experiments are performed on the SSP2000 dataset. For a fair comparison, all the experiments are implemented on the PyTorch framework in Ubuntu 20.04 + Python 3.8 environment with one NVIDIA GeForce RTX 3090 GPU (24GB memory). We utilize the stochastic gradient descent optimizer to train our network. The network needs to be trained for 200 epochs, with the initial learning rate set at 0.025, momentum at 0.9, and weight decay at 0.0001. The optimization process of the learning rate includes a warmup phase and reaches 0.025 at the 100th epoch, which helps improve the stability of training. Based on testing, the SRPCNet needs to be trained for approximately 5.7 h.




(a) (b)

Fig. 11. (a) F-measure and (b) PR curves of different methods on the SSP2000 dataset.

### B. Evaluation Metrics

We adopt eight evaluation metrics to quantitatively measure the experimental performance, which include $S$-measure ($S_\alpha$), adaptive and maximum $E$-measure ($E_\xi^{\text{adp}}$ and $E_\xi^{\max}$), adaptive, maximum, mean, and weighted $F$-measure ($F_\beta^{\text{adp}}$, $F_\beta^{\max}$, $F_\beta^{\text{mean}}$, and $F_\beta^{w}$) and mean absolute error (MAE).

### C. Comparison Experiment

Eight state-of-the-art SOD methods and seven state-of-the-art COD methods are used for comparison with our SRPCNet. SOD methods include A3Net [17], VST [29], BBRF [30], SelfReformer [31], TCGNet [32], CSEPNet [16], ICON [18], and EDRNet [15], and COD methods include PRNet [21], DTINet [33], FRINet [34], MRRNet [35], FEDER [22], HitNet [23], and LSR [36]. The results of all methods are generated by public source codes under the default parameter settings.

*1) Quantitative Results:* Quantitative results in Table II shows that when dealing with ISDs images of SSPs, which have challenges, such as MCNO and CMO, the proposed SRPCNet outperforms other compared methods across all eight evaluation metrics, which clearly demonstrates the advantages of the proposed SRPCNet. Notably, our method exhibits significant improvement in all four $F$-measures. Furthermore, the MAE values for all methods are relatively small, which is attributed to the small size of the objects in the dataset. In addition, to evaluate our method more intuitively, we also draw the F-measure and precision–recall (PR) curves, which are shown in Fig. 11(c) and (d).

*2) Qualitative Results:* Fig. 12 presents the visual results of our method and some representative methods. The results in the red box demonstrate that SRPCNet can detect COs and small objects effectively. The results in the blue box show that when images contain the CMO challenge and the MCNO challenge, our method can adequately focus on objects of different scales as well as COs. The results in the green box illustrate that for DDs, our method can provide more detailed information and is capable of detecting defects more comprehensively. In addition, we present the visualization of detection results for all defect categories in the SSP2000 dataset in Fig. 13, further

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TABLE II
QUANTITATIVE COMPARISONS WITH 15 STATE-OF-THE-ART METHODS ON THE SSP2000 DATASET

| Methods | Year | Type | Backbone | Crop size | FLOPs (G)↓ | Params (M)↓ | Evaluation Metrics | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | $S_\alpha \uparrow$ | $E_\xi^{adp} \uparrow$ | $E_\xi^{max} \uparrow$ | $F_\beta^{adp} \uparrow$ | $F_\beta^{max} \uparrow$ | $F_\beta^{mean} \uparrow$ | $F_\beta^{w} \uparrow$ | MAE↓ |
| EDRNet [15] | 2020 | SOD | ResNet | $224^2$ | 32.4 | 39.3 | 0.8025 | 0.8436 | 0.9357 | 0.5488 | 0.7098 | 0.6561 | 0.6383 | 0.0028 |
| VST [29] | 2021 | SOD | T2T-Vit | $224^2$ | 23.2 | 44.1 | 0.7925 | 0.5925 | 0.9323 | 0.3295 | 0.7265 | 0.6080 | 0.5769 | 0.0032 |
| ICON [18] | 2022 | SOD | Swin-B | $384^2$ | 52.8 | 92.2 | 0.8089 | 0.9128 | 0.9514 | 0.6457 | 0.7358 | 0.7110 | 0.7165 | **0.0024** |
| CSEPNet [16] | 2022 | SOD | VGG16 | $224^2$ | 45.4 | **18.8** | 0.8029 | **0.9408** | 0.9489 | 0.6654 | 0.7151 | 0.6954 | 0.7069 | 0.0026 |
| A3Net [17] | 2023 | SOD | Swin-B | $384^2$ | 48.4 | 89.7 | 0.8131 | 0.9305 | **0.9544** | 0.6628 | 0.7269 | 0.7077 | 0.7159 | **0.0024** |
| BBRF [30] | 2023 | SOD | Swin-T | $352^2$ | 46.4 | 74.4 | **0.8215** | **0.9393** | 0.9474 | **0.7027** | **0.7465** | **0.7230** | **0.7320** | **0.0024** |
| SelfReformer [31] | 2023 | SOD | PVTv2 | $224^2$ | 12.8 | 90.7 | 0.7671 | 0.6186 | 0.9324 | 0.3444 | 0.6643 | 0.5852 | 0.5632 | 0.0034 |
| TCGNet [32] | 2023 | SOD | HRNet | $224^2$ | 20.5 | 70.3 | 0.8174 | 0.7937 | 0.9477 | 0.5183 | 0.7297 | 0.6644 | 0.6670 | 0.0029 |
| DTINet [33] | 2023 | COD | MiT-B5 | $256^2$ | 145.9 | 264.8 | 0.7137 | 0.6512 | 0.8850 | 0.3612 | 0.5179 | 0.4892 | 0.4978 | 0.0045 |
| FRINet [34] | 2023 | COD | ViT | $384^2$ | 96.6 | 114.6 | 0.7477 | 0.8653 | 0.9310 | 0.5162 | 0.6219 | 0.5939 | 0.5898 | 0.0032 |
| MRRNet [35] | 2023 | COD | ResNet | $384^2$ | 24.8 | 87.6 | 0.7802 | 0.7850 | 0.9423 | 0.5111 | 0.7031 | 0.6369 | 0.6476 | 0.0032 |
| FEDER [22] | 2023 | COD | Res2Net | $384^2$ | 36.0 | 42.1 | 0.7791 | 0.7652 | 0.9367 | 0.4947 | 0.7010 | 0.6179 | 0.6311 | 0.0030 |
| HitNet [23] | 2023 | COD | PVTv2 | $384^2$ | 16.8 | 25.7 | 0.8114 | 0.9166 | 0.9422 | **0.6671** | 0.7189 | 0.6924 | 0.7061 | 0.0027 |
| LSR [36] | 2023 | COD | ResNet | $384^2$ | 20.8 | 28.7 | 0.7509 | 0.6502 | 0.9417 | 0.3964 | 0.7030 | 0.5974 | 0.6078 | 0.0034 |
| PRNet [21] | 2024 | COD | SMT | $384^2$ | 10.2 | **12.7** | **0.8314** | 0.8787 | **0.9558** | 0.6261 | **0.7689** | **0.7194** | **0.7312** | **0.0022** |
| SRPCNet(Ours) | 2024 | SOD | VGG16 | $224^2$ | 50.0 | **18.1** | **0.8441** | **0.9443** | **0.9561** | **0.7283** | **0.7914** | **0.7798** | **0.7698** | **0.0021** |

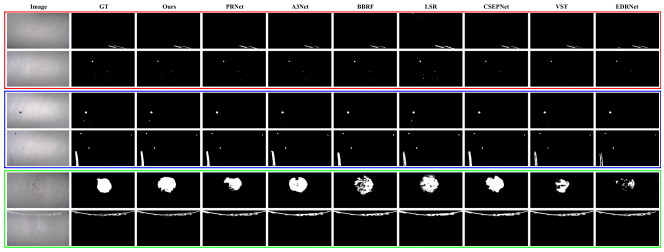



Fig. 12. Visualization comparison on the SSP2000 dataset. Please zoom in for the best view.

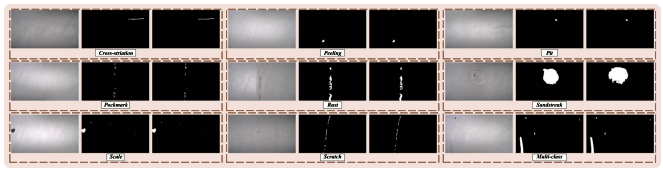



Fig. 13. Visualization of detection results for all categories of defects in the SSP2000 dataset. Please zoom in for the best view.

TABLE III
RESULTS OF THE ABLATION STUDIES

| No. | Modules | | | Evaluation metrics | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | SPA | HAIM | BSFM | $S_\alpha \uparrow$ | $E_\xi^{adp} \uparrow$ | $E_\xi^{max} \uparrow$ | $F_\beta^{adp} \uparrow$ | $F_\beta^{max} \uparrow$ | $F_\beta^{mean} \uparrow$ | $F_\beta^{w} \uparrow$ | MAE↓ |
| 1 | | | | 0.6936 | 0.7089 | 0.7622 | 0.3941 | 0.5367 | 0.5258 | 0.4924 | 0.0053 |
| 2 | ✓ | | | 0.7791 | 0.8071 | 0.9106 | 0.5281 | 0.7231 | 0.7000 | 0.6758 | 0.0033 |
| 3 | | ✓ | | 0.8263 | 0.9172 | 0.9475 | 0.6868 | 0.7811 | 0.7647 | 0.7519 | 0.0023 |
| 4 | | | ✓ | 0.8378 | 0.9301 | 0.9503 | 0.7069 | 0.7855 | 0.7713 | 0.7621 | 0.0021 |
| 5 | ✓ | ✓ | | 0.8432 | 0.9354 | 0.9514 | 0.7200 | 0.7904 | 0.7728 | 0.7676 | 0.0022 |
| 6 | | ✓ | ✓ | 0.8426 | 0.9352 | 0.9515 | 0.7162 | 0.7875 | 0.7728 | 0.7646 | 0.0021 |
| 7 | ✓ | | ✓ | 0.8425 | 0.9371 | 0.9532 | 0.7223 | 0.7907 | 0.778 | 0.7685 | 0.0021 |
| 8 | ✓ | ✓ | ✓ | **0.8441** | **0.9443** | **0.9561** | **0.7283** | **0.7914** | **0.7798** | **0.7698** | **0.0021** |

demonstrating the ability of SRPCNet to detect various types of ISDs in SSPs.

## D. Ablation Studies

We use the same eight evaluation metrics and conduct a series of ablation experiments on the SSP2000 dataset to investigate the contributions of the crucial modules quantitatively. The quantitative results of the ablation studies are presented in Tables III and IV. Notably, we remove all the tricks used in SRPCNet, obtaining the baseline results, as shown in Table III No. 1.

*1) Impact of different components:* To demonstrate the effectiveness of different modules in SRPCNet, we separately use

TABLE IV
RESULTS OF THE EFFECTIVENESS VERIFICATION OF SPA AND HAIM

| No. | Modules | | | | | | Evaluation metrics | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SPA | HAIM | BSFM | MFAM | DFA | ERF | $S_\alpha \uparrow$ | $E_\xi^{adp} \uparrow$ | $E_\xi^{max} \uparrow$ | $F_\beta^{adp} \uparrow$ | $F_\beta^{max} \uparrow$ | $F_\beta^{mean} \uparrow$ | $F_\beta^{w} \uparrow$ | MAE↓ |
| 1 | | ✓ | ✓ | | | | 0.8425 | 0.9371 | 0.9532 | 0.7223 | 0.7907 | 0.7780 | 0.7685 | **0.0021** |
| 2 | | ✓ | ✓ | ✓ | | | 0.8393 | 0.9369 | **0.9563** | 0.7161 | 0.7837 | 0.7713 | 0.7637 | **0.0021** |
| 3 | | ✓ | ✓ | | ✓ | | 0.8418 | 0.9371 | 0.9506 | 0.7188 | 0.7848 | 0.7729 | 0.7618 | **0.0021** |
| 4 | | ✓ | ✓ | | | ✓ | 0.8396 | 0.9358 | 0.9510 | 0.7207 | 0.7884 | 0.7759 | 0.7660 | 0.0022 |
| 5 | ✓ | ✓ | ✓ | | | | **0.8441** | **0.9443** | 0.9561 | **0.7283** | **0.7914** | **0.7798** | **0.7698** | **0.0021** |
| No. | SPA | HAIM | BSFM | FPN | PA | BiFPN | $S_\alpha \uparrow$ | $E_\xi^{adp} \uparrow$ | $E_\xi^{max} \uparrow$ | $F_\beta^{adp} \uparrow$ | $F_\beta^{max} \uparrow$ | $F_\beta^{mean} \uparrow$ | $F_\beta^{w} \uparrow$ | MAE↓ |
| 6 | ✓ | | ✓ | | | | 0.8426 | 0.9352 | 0.9515 | 0.7162 | 0.7875 | 0.7728 | 0.7646 | **0.0021** |
| 7 | ✓ | | ✓ | ✓ | | | 0.8428 | 0.9382 | 0.9545 | 0.7160 | 0.7891 | 0.7721 | 0.7652 | **0.0021** |
| 8 | ✓ | | ✓ | | ✓ | | 0.8339 | 0.9402 | 0.9488 | 0.7215 | 0.7826 | 0.7697 | 0.7587 | 0.0022 |
| 9 | ✓ | | ✓ | | | ✓ | 0.8400 | 0.9296 | 0.9479 | 0.7073 | 0.7810 | 0.7677 | 0.7583 | 0.0022 |
| 10 | ✓ | ✓ | ✓ | | | | **0.8441** | **0.9443** | **0.9561** | **0.7283** | **0.7914** | **0.7798** | **0.7698** | **0.0021** |

the three modules (No. 2, No. 3, and No. 4). Compared with the baseline results, these tests confirm each module can improve the detection accuracy, which is attributed to SPA enhancing perceptual capabilities, HAIM effectively concentrating and integrating multilayer feature information, and BSFM sufficiently enhancing and extracting features. In addition, we also conduct experiments on different combinations of two modules (No. 5, No. 6, and No. 7) to evaluate the impact when two modules are used together. Compared with using a single module, the results demonstrate that each combination can improve the performance, and the best detection results achieve when using SPA, HAIM, and BSFM modules together.

*2) Effectiveness of SPA and HAIM:* We conduct more ablation experiments to further demonstrate the effectiveness of SPA and HAIM. We removed SPA and replaced it with several advanced feature enhancement modules, such as MFAM from [37], DFA from [18], and ERF from [38]. In addition, we removed HAIM and its corresponding feature fusion structure, replacing it with several well-known feature fusion structures, including FPN from [39], path aggregation from [40], and BiFPN from [41]. The quantitative results obtained are presented in Table IV. As shown by the experimental results, the superior performance of SPA and HAIM, attributed to the effective perceptual capability of SPA and the comprehensive feature fusion of HAIM, outperforms other methods.

## E. Generalization Experiment

To further validate the generalization ability of the SRPCNet, we conduct generalization experiments on the SD-saliency-900 dataset [42]. Following the experimental setup of EDRNet [15], the dataset is divided into a training set with 540 images and a testing set with 360 images. We compare SRPCNet with fifteen state-of-the-art methods, including ten SOD methods (EDRNet [15], VST [29], ICON [18], BBRF [30], MEANet [43], SelReformer [31], TCGNet [32],

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TABLE V
QUANTITATIVE COMPARISONS ON THE SD-SALIENCY-900 DATASET

| Methods | Year | Type | Backbone | Evaluation Metrics | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | $S_\alpha \uparrow$ | $E_\xi^{adp} \uparrow$ | $E_\xi^{max} \uparrow$ | $F_\beta^{adp} \uparrow$ | $F_\beta^{max} \uparrow$ | $F_\beta^{mean} \uparrow$ | $F_\beta^w \uparrow$ | MAE↓ |
| EDRNet [15] | 2020 | SOD | ResNet | 0.8759 | 0.9517 | 0.9601 | 0.8323 | 0.8672 | 0.8517 | 0.8457 | 0.0252 |
| VST [29] | 2021 | SOD | T2T-Vit | 0.8708 | 0.9237 | 0.9590 | 0.7812 | 0.8589 | 0.8228 | 0.8147 | 0.0277 |
| ICON [18] | 2022 | SOD | Swin-B | 0.8760 | **0.9518** | 0.9639 | 0.8317 | 0.8799 | 0.8498 | 0.8483 | **0.0236** |
| BBRF [30] | 2023 | SOD | Swin-T | 0.8769 | 0.9580 | 0.9608 | **0.8563** | 0.8766 | **0.8587** | 0.8590 | 0.0219 |
| MEANet [43] | 2023 | SOD | MobileNetV2 | **0.8787** | 0.9566 | 0.9605 | **0.8503** | **0.8807** | 0.8589 | 0.8564 | 0.0242 |
| SelfReformer [31] | 2023 | SOD | PVTv2 | 0.8720 | 0.9505 | **0.9631** | 0.8251 | 0.8730 | 0.8492 | 0.8475 | 0.0241 |
| TCGNet [32] | 2023 | SOD | HRNet | 0.8746 | 0.9491 | 0.9635 | 0.8252 | 0.8756 | 0.8490 | 0.8480 | 0.0238 |
| DTINet [33] | 2023 | COD | MiT-B5 | 0.8698 | 0.9447 | 0.9576 | 0.8222 | 0.8611 | 0.8411 | 0.8355 | 0.0251 |
| MRRNet [35] | 2023 | COD | ResNet | 0.8782 | 0.9413 | 0.9621 | 0.8228 | 0.8774 | 0.8444 | 0.8401 | 0.0247 |
| FEDER [22] | 2023 | COD | Res2Net | 0.8684 | 0.9409 | 0.9593 | 0.8139 | 0.8711 | 0.8344 | 0.8327 | 0.0254 |
| LSR [36] | 2023 | COD | ResNet | 0.8667 | 0.9302 | 0.9583 | 0.7974 | 0.8645 | 0.8277 | 0.8180 | 0.0274 |
| PRNet [21] | 2024 | COD | SMT | 0.8792 | 0.9457 | 0.9617 | 0.8391 | 0.8826 | 0.8528 | **0.8527** | 0.0233 |
| ADMNet [44] | 2024 | SOD | ADMNet | 0.8654 | 0.9359 | 0.9553 | 0.7973 | 0.8545 | 0.8276 | 0.8225 | 0.0290 |
| MINet [45] | 2024 | SOD | MI-based | 0.8643 | 0.9367 | 0.9573 | 0.7972 | 0.8560 | 0.8289 | 0.8220 | 0.0278 |
| DASOD [46] | 2024 | SOD | ResNet | 0.8682 | 0.9378 | 0.9589 | 0.8093 | 0.8653 | 0.8323 | 0.8294 | 0.0257 |
| SRPCNet(Ours) | 2024 | SOD | VGG16 | 0.8840 | 0.9580 | 0.9635 | 0.8593 | 0.8855 | 0.8677 | 0.8600 | 0.0237 |

TABLE VI
PERFORMANCE OF THE PROPOSED SYSTEM

| Performance metrics | Value |
|---|---|
| Detectable Inner Diameter Range (mm) | 100-150 |
| Detection Speed (m/min) | 6 |
| Weight of PASS (Kg) | 2.5 |
| Length of PASS (m) | 0.8 |
| Single Use Duration (h) | 1.5 |
| Charging Time (h) | 0.5 |
| Params of SRPCNet (M) | 18.1 |
| Model Size of SRPCNet (MB) | 69.5 |
| Speed of SRPCNet (FPS) | 22.2 |

ADMNet [44], MINet [45], DASOD [46]) and five COD methods (DTINet [33], MRRNet [35], FEDER [22], LSR [36], PRNet [21]). We use the same eight evaluation metrics to quantitatively evaluate the detection performance, as shown in Table V. The results indicate that SRPCNet also achieves satisfactory performance on the SD-saliency-900 dataset, effectively demonstrating the generalization ability and effectiveness of SRPCNet. These results further confirm that SRPCNet is capable of handling other types of steel surface defect detection tasks.

### F. Practical Application

The proposed detection system achieves automated detection for SSPs with inner diameters ranging from 100 to 150 mm. More importantly, the highly visualized detection results showed in the IVS make it possible for inspection workers to observe defects. With the visualized detection results, the workers can further analyze the causes of defects and adjust production processes accordingly. The result also provides guidance for the next grinding stage. Extensive testing experiments in the SSPs factory shows that the proposed detection system achieves the detection metrics outlined in Table VI. Specifically, after the PASS robot enters the pipeline, it moves forward at a constant speed of 0.4 m/s, while each visual inspection unit captures images at a rate of 13 fps. As a result, the five visual inspection units collectively capture 65 images per second. For an SSPs with a length of 6 m, the PASS needs to travel for 15 s, during which 975 images are collected. These images are then compressed and wirelessly transmitted to the server, with the transmission time can be negligible. Given that the processing speed of SRPCNet is 22.2 fps, processing these images takes approximately 44 s. Therefore, the total time required to detect a 6 m long SSP is 1 min. Combining with the quantitative experiment results, it demonstrates the detection system can accurately and efficiently detect ISDs in SSPs. Our application demonstration video is available at.[1]

### G. Limitation Analysis and Further Discussion

Although the proposed detection system can achieve efficient, accurate, and highly visualized detection, there is still improvement space in overall detection efficiency. Specifically, the current detection flow needs the server to process all the captured images. However, the development boards in the PASS also have computational capabilities that can quickly filter most of the defect-free images and then send the remaining images to the server for further high-precision detection. In the future, we plan to deploy filter algorithms on embedded development boards, which will efficiently shorten the detection time, thereby enhancing detection efficiency.

Moreover, through experiments and observations in the SSPs factory, we find that lots of SSPs awaiting internal surface detection are in the piled state, highlighting the inefficiency of current methods and verifying the necessity of the proposed detection system. As we move toward the era of intelligent factories, each production stage should aim for high automation and continuity between each other. Although the proposed detection system can effectively detect SSPs in the piled state, it still requires manual insertion and removal of the PASS into and out of the SSPs. Therefore, we will strive to develop a new ISD detection system to achieve highly automated, assembly line-style detection.

## VI. CONCLUSION

In this article, we construct an detection system for ISDs in SSPs, which includes the PASS and the IVS. The system achieves accurate and efficient detection of ISDs in SSPs. What is more, we construct the first dataset for ISDs in SSPs (SSP2000). The SSP2000 dataset not only fills the gap for ISDs in SSPs, but also provides a reference for the inspection stage in the production process of SSPs. In addition, the proposed SRPCNet effectively addresses the challenges in the SSP2000 and achieves high detection performance. Meanwhile, the visualized detection results in the IVS can effectively guide the grinding process, improving the existing production workflow. In the future, we will continue enhancing the detection capabilities of the system and developing more automated internal surface visual detection systems.

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